

↑ Game Playing with AI

- ① Game playing is non-trivial
- ↳ needs "human like" intelligence
 - ↳ can be very complex
 - ↳ decision making within limited time
- but,

- ② Games are,
- ↳ well-defined & Repeatable
 - ↳ limited & accessible

↑ Game Playing Algorithms

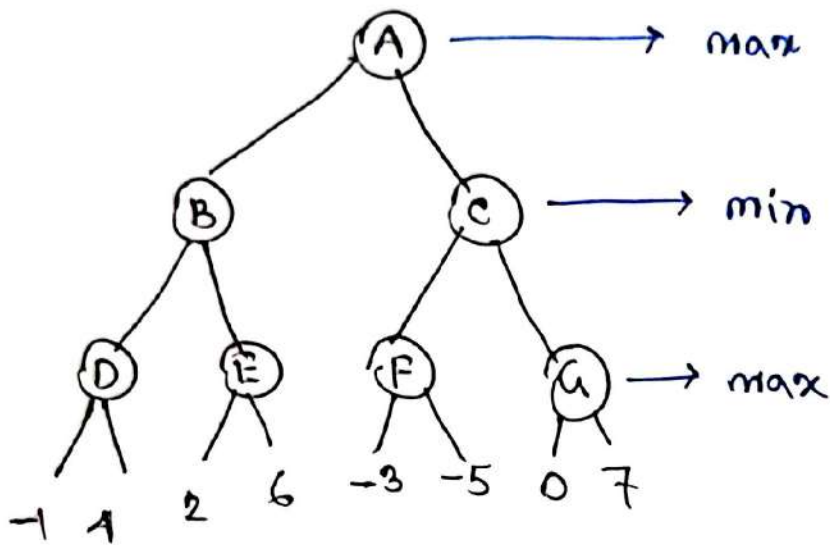
- ① Mini-max Algorithm ② ~~the~~ $\alpha\beta$ Algorithm

- ↳ maximize utility
- ↳ minimize utility



Shayan / 2

Minimax Algorithm

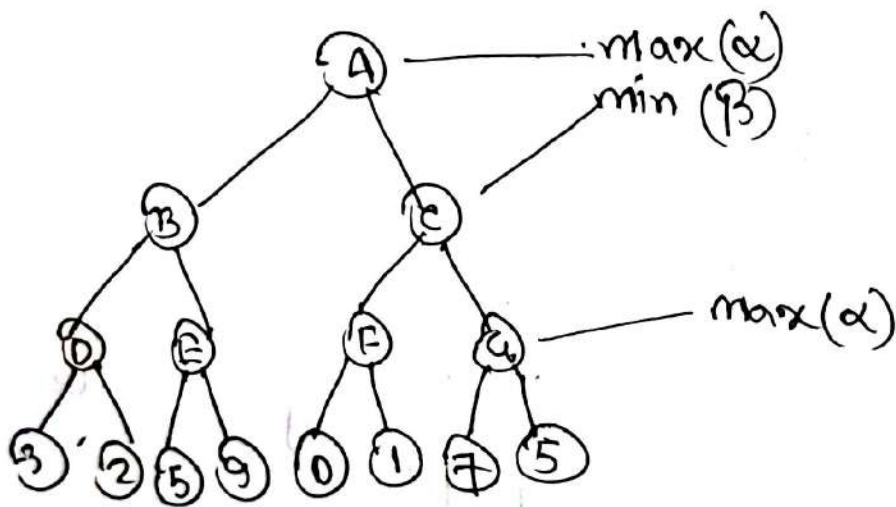


DFS Approach

Top level Depth 2
খাতি গুরুত্বপূর্ণ হবে
গুরুত্বপূর্ণ

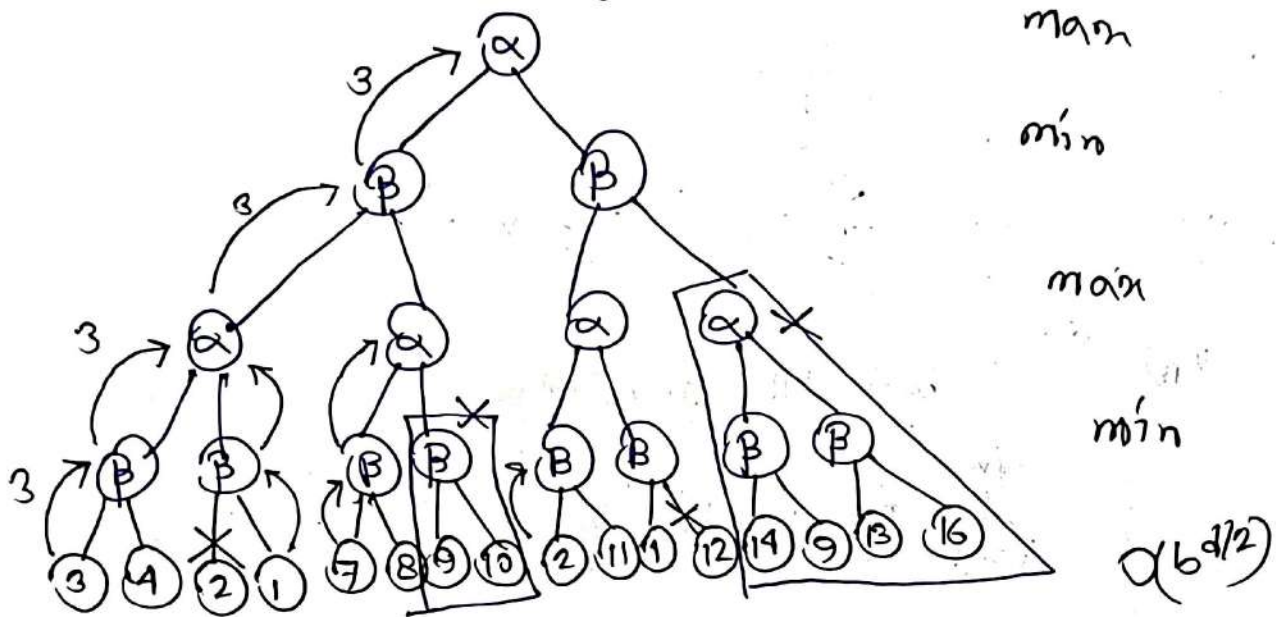
time complexity
 $O(b^d)$

Alpha-Beta Pruning



Shayaro | ③

→ Alpha-Beta Pruning



⇒ Graph

⇒ Pruning justifications

~~→ New~~

→ Neural Networks: Geeks for Geeks

① Neurons

② Adjustable weights

$$f(\sum W_i x_i) = y \rightarrow f(\sum W_i x_i + \text{bias value}) = y = \hat{y}$$

③ Back Propagation

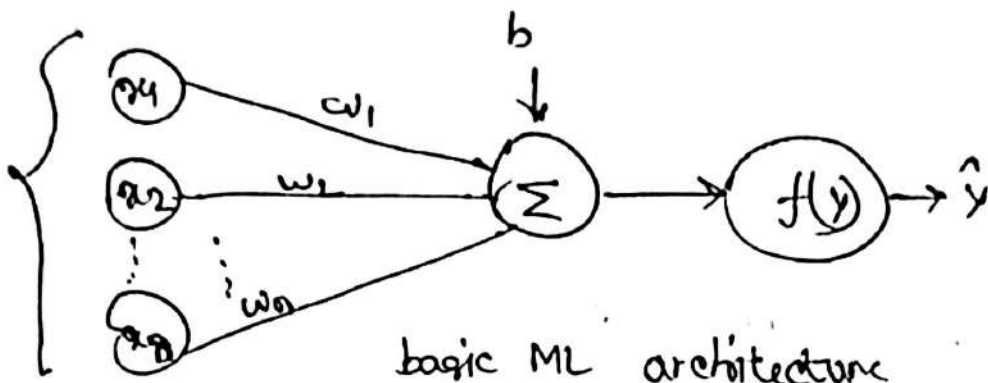
④ Learning weight ***

Types of Neural Networks

"Attention is all you need" paper → overfitting

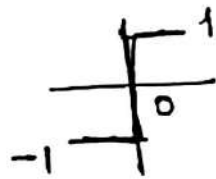
Early Stopping

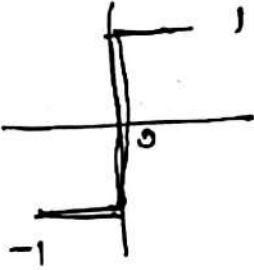
Shahin



$$f \left\{ \sum_{i=1}^n x_i w_i + b \right\}$$

Sgn function
-1, 0, 1



$$\text{sgn}(y) = \begin{cases} +1 & y > 0 \\ 0 & y = 0 \\ -1 & y < 0 \end{cases}$$


$$\hat{y} = y = \text{sgn}(y)$$

if $y > 0$ then

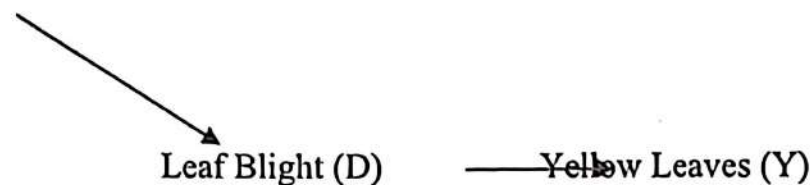
output is 1

1.

A farmer is monitoring a tomato field for a fungal disease called **Leaf Blight (D)**. Two factors influence whether the disease occurs: **High Humidity (H)** and **Poor Soil Drainage (S)**. If the disease is present, the plant is more likely to show **Yellow Leaves (Y)**. The Bayesian Network is shown below:

High Humidity (H)

Poor Soil Drainage (S)



Variable	State	Probability
High Humidity (H)	True	0.60
	False	0.40
Poor Soil Drainage (S)	True	0.30
	False	0.70

(H)	(S)	P(D=True)
True	True	0.90
True	False	0.70
False	True	0.60
False	False	0.10

(D)	P(Y=True)
True	0.85
False	0.20

- Find the joint probability that: High Humidity is present, Poor Soil Drainage is present, Leaf Blight occurs, and Yellow Leaves are observed.
- Find the probability that **Yellow Leaves** are observed.

2

Sample	Weather	Temperature	Windy	Play Outside
1	Sunny	Hot	No	Yes
2	Sunny	Hot	Yes	No
3	Cloudy	Mild	No	Yes
4	(Rainy)	(Cool)	(No)	Yes
5	(Rainy)	(Cool)	Yes	No
6	Cloudy	Cool	Yes	Yes
7	Sunny	Mild	No	Yes
8	(Rainy)	Mild	No	Yes

4

- a) Calculate the prior probabilities of the classes Play Outside = Yes and Play Outside = No.
- b) Using the Naïve Bayes classifier, compute the posterior probabilities:
 - I. $P(\text{Yes}|\text{Rainy}, \text{Cool}, \text{No})$
 - II. $P(\text{No}|\text{Rainy}, \text{Cool}, \text{No})$
- c) Based on your calculations, predict whether the children will play outside. Justify your answer by comparing the posterior probabilities.